Can open data beat the operator? A paired comparison of a reproducible arrival prediction method with the official real-time waiting times of an interurban bus network

Yeray Garrido[1]

Abstract

Passengers of many bus networks receive real-time waiting times from the operator, either directly or through apps that relay them, but the accuracy of these predictions is rarely compared with alternatives on the same buses. Bizkaibus, the interurban bus network of Biscay (Spain), shows its own waiting times in an official app and publishes, as open data, the positions of its vehicles and its timetable. For one week in November 2026 we recorded the official waiting times of 60 stops every minute and the live positions every 20 seconds, and compared, for the same bus at the same stop and minute, the official announcement with the arrival predicted by a reproducible method that uses only the open data, both against the actual passage reconstructed from the feed. Over 160,046 paired predictions on five normal days, both are accurate when the bus is less than three minutes away (86% and 88% within two minutes). Further ahead, the method is far more accurate: with the bus 6 to 10 minutes away, 71% of its predictions are within two minutes against 37% of the official ones, and 61% against 27% at 10 to 20 minutes. The official errors are mostly late announcements: in about half of the announcements made 10 to 30 minutes ahead, the bus arrives more than two minutes earlier than announced. The capture program, the data and the analysis are openly available.

Keywords: bus arrival prediction, real-time passenger information, operator predictions, paired evaluation, SIRI, GTFS, open data, reproducibility

[1]Independent researcher, Bilbao, Biscay, Spain. ORCID: 0009-0008-0459-0294. E-mail: garridoparrayeraytx@gmail.com. Code and data: `https://github.com/Garridoparrayeray/bide-eta`.

1. Introduction

Real-time arrival information shortens the time passengers wait at stops and makes that wait less uncertain (K. E. Watkins et al. 2011; Brakewood and K. Watkins 2019), but only if it is accurate: inaccurate predictions erode trust and can make passengers miss the bus they were told to expect (Gooze et al. 2013; Rahman et al. 2013). Most passengers receive these predictions from the operator itself, through its own app and stop displays, or from third-party apps that relay the operator's figures. How accurate the operator's predictions are, and whether a third party with only the operator's open data can do better, is rarely measured on the same buses and at the same moments.

This paper makes that measurement for Bizkaibus, the interurban bus network of the province of Biscay (Spain). Since 2023 Bizkaibus shows its own real-time waiting times in an official app and on some stop displays. It also publishes, as open data, the GPS position of every vehicle in service (a SIRI Vehicle Monitoring feed, CEN 2022) and its timetable (GTFS, MobilityData 2026), but not the predictions themselves. A companion paper (Garrido 2026b) and a technical report (Garrido 2026a) describe and evaluate a method that predicts arrivals from those two open sources alone, used by the independent Bizkaibus+ app. That evaluation compared the method with the published timetable, which is the obvious but weak baseline: after the first hour of a trip, most Bizkaibus buses pass a stop more than two minutes before the published time.

Here the baseline is the operator's own live prediction. For one week in November 2026 we recorded, every minute, the waiting times that the official service gave for 60 stops, together with the live feed every 20 seconds. We then compared, for the same bus at the same stop and the same minute, the waiting time announced by the operator with the arrival predicted by the method, and both with the moment the bus actually passed.

The contributions are:

1. A paired protocol to compare a third-party predictor with an operator's real-time predictions when the operator publishes positions but not predictions: what is captured, how each announcement is matched to a vehicle, how the actual passage is reconstructed and how the comparison avoids giving either side an advantage.
2. The comparison itself, over one week of data, by how far ahead the bus is, by type of day and by type of stop, with confidence intervals that respect the dependence between predictions of the same trip.

3. The data, the capture program and the analysis, openly available, so that the comparison can be repeated on other days or other networks.

Section 2 reviews related work, Section 3 describes the data, Section 4 the comparison, Section 5 the results and Section 6 their interpretation and limits.

2. Related work

The accuracy of operator real-time information has been studied mostly through the operator's own logs. Cats and Loutos (2016) evaluated the bus arrival information system of Stockholm against automatic vehicle location records and found that prediction errors grow with the prediction horizon and depend on the type of stop and on the time of day. Studies of passenger benefits (K. E. Watkins et al. 2011; Mishalani et al. 2006; Liu and Miller 2020) use the operator's predictions as given and show that the value of real-time information depends on its accuracy (Gooze et al. 2013). On the other side, third-party predictors built only on public data have been shown to be feasible: García-Mauriño et al. (2024) improve arrival predictions using only data available through public transport APIs, and Wessel et al. (2017) reconstruct actual passage times from a public vehicle location feed, the same kind of ground truth used here. The gap that this paper addresses is the direct comparison: operator predictions and a third-party predictor on the same vehicles, stops and moments, against the same actual passage, with the operator's predictions recorded as passengers saw them.

The evaluation follows the rules of out-of-sample forecasting evaluation (Tashman 2000): the method is calibrated only with data from before the week that is measured, so that nothing from the test days leaks into it (Kaufman et al. 2012).

3. Data

3.1. What was captured

A program running on GitHub Actions captured, from 06:00 to 23:00 Madrid time (the night feed repeats stale data):

- **The live feed**, every 20 seconds: the public SIRI Vehicle Monitoring file of Bizkaibus, with the GPS position, the trip and the next stop of every vehicle in service. The file changes every 60 seconds, so two of every three copies repeat the previous one.

- **The official waiting times**, every minute, for each of 60 stops: the response of the web service behind the official Bizkaibus app (`GetPasoParadaMobile_JSON`). For each line that serves the stop, the service gives the next two buses (here e_1 and e_2), each with its waiting time in whole minutes and its distance in metres. The stops are queried one after another, so each stop is queried about once a minute.
- **The timetable** of the day (GTFS), once a day.

The 60 stops are intermediate stops with many buses and lines, at least 15 minutes after the start of the trip (where predictions are hardest and most useful), plus 28 *priority* stops: stops pointed out by passengers and the Bilbao stops where the actual passage deviates most from the published time. Each stop and the reason for its choice are listed in the repository (`paradas.json`).

3.2. Days

The week runs from Sunday 8 to Saturday 14 November 2026 (Table 1). Wednesday 11 was a general strike: Bizkaibus ran minimum services (about 30%), with about 160 vehicles in the feed against 240 on a normal day. That day is analysed separately and does not enter the results of the week. Three days are incomplete: on Monday 9 the morning block was lost because the capture overran its time limit; on Thursday 12 the official service stopped answering over plain HTTP at 17:12 (it answers over HTTPS, which the capture uses from then on), so that afternoon has no official times; and on Saturday 14 the capture started at 09:41. The live feed has no gap longer than two minutes on any day.

4. Comparison design

4.1. The two predictions

Official. For a query at time t to stop s, an announcement of m minutes for a bus of line ℓ gives the predicted arrival $\hat{a}_{\text{off}} = t + 60\,m$.

Method. The method of Garrido (2026b) is recomputed from the captured feed with its open reference implementation, which reproduces the production engine exactly (it passes the 407 test vectors produced by that engine). It is not taken from the live app, so that anyone can repeat it. Its learned tables (segment times, pace factors and per-trip corrections) are those of Saturday 7 November, calibrated only with data from before the week, and they are not updated during the week. For the query at time t, the

Table 1: Days of the capture (Madrid time). The feed and the official service were captured from about 06:00 to 23:00, except as noted.

Date	Day	Coverage
8 Nov 2026	Sunday	Complete
9 Nov 2026	Monday	12:42–23:00 (morning block lost: the capture overran its time limit)
10 Nov 2026	Tuesday	Complete
11 Nov 2026	Wednesday	Complete, *analysed separately*: general strike, minimum services (about 30%)
12 Nov 2026	Thursday	Feed complete; official times until 17:12 (the service stopped answering over HTTP)
13 Nov 2026	Friday	Complete
14 Nov 2026	Saturday	From 09:41

method's prediction is the one it gave in the last copy of the feed before t, provided that copy is at most 60 seconds old; it is therefore never more recent than the official one. Its predicted arrival $\hat{a}_{\mathrm{met}}$ is compared in seconds and also rounded to whole minutes from t, as a passenger would see it: $\hat{a}_{\mathrm{met,min}} = t + 60\,\max\bigl(0, \mathrm{round}\bigl((\hat{a}_{\mathrm{met}} - t)/60\bigr)\bigr)$. The main comparison uses the rounded value, so that neither side gains from the resolution of its figures.

4.2. The actual passage

The actual passage of a bus at a stop, a, is reconstructed from the feed as in the companion paper: it is the midpoint between the last copy of the feed in which the bus had not yet passed the stop (its next stop is the stop itself or an earlier one) and the first copy in which it had, without the bus changing trip in between. Because the feed changes every 60 seconds, a is known to within about ± 30 seconds. The same a is used for both predictions, so this uncertainty adds the same noise to both errors and does not favour either.

4.3. Matching announcements to buses

The official service does not say which vehicle it announces. For a query at t to stop s, its e_1 for line ℓ is matched to the first bus of line ℓ that actually passes s after t, and its e_2 to the second. This is the bus a passenger waiting at s would take. Lines listed twice in the same response (two routes of the same line) are left out, as are announcements whose matched bus passes more than 30 minutes after t or is not tracked by the feed. Because buses of the same line can bunch, the results are repeated without the pairs in which

Table 2: Errors by time to the actual passage, five normal days. *Median*: median absolute error (s). *±2 min*: share of predictions within two minutes (%). *Early*: share of buses that arrive more than two minutes *before* the announced time (%), the error that makes passengers miss the bus. The method is rounded to whole minutes, as shown to passengers.

		Official			Method		
Minutes to arrival	Pairs	Median	±2 min	Early	Median	±2 min	Early
0–3	19,329	43	86	14	38	88	9
3–6	18,973	114	52	27	56	77	11
6–10	24,119	167	37	39	68	71	13
10–20	53,865	246	27	51	91	61	16
20–30	43,760	354	19	52	128	48	16
All	160,046	191	36	42	81	64	14

another bus of the same line passes within two minutes of the matched one; the conclusions do not change (Section 5).

4.4. Metrics

The error of each prediction is $\hat{a} - a$ (positive: the bus arrives earlier than announced). For each source we give the median absolute error, the share of predictions within ±1 and ±2 minutes, and the median signed error, by horizon $a - t$ (0–3, 3–6, 6–10, 10–20 and 20–30 minutes), by type of day and by type of stop. Predictions of the same trip are not independent, so confidence intervals for the difference between the two sources are obtained by bootstrap over whole trips (Efron and Tibshirani 1993): each of 400 resamples draws trips with replacement, with all their pairs.

5. Results

5.1. Overall

Over the five normal days analysed so far (8, 9, 10, 12 and 13 November), 160,046 official announcements were paired with a bus and with the method's prediction for the same bus at the same moment: 2,854 passages of 879 trips of 72 lines at the 60 stops. Table 2 gives the errors by how far away the bus was.

When the bus is less than three minutes away, both are accurate and close: 86% of official announcements and 88% of the method's predictions are within two minutes. From three minutes on, they separate. With 6 to 10 minutes to go, 37% of official announcements are within two minutes against 71% for the method, and the median error is 167 s against 68 s.

With 10 to 20 minutes to go, the figures are 27% and 61%. The difference in the share within two minutes, method minus official, has 95% bootstrap intervals (by trip) of $[+1.2, +4.0]$ points at 0–3 minutes, $[+22.7, +26.9]$ at 3–6, $[+32.0, +36.9]$ at 6–10, $[+31.8, +36.3]$ at 10–20 and $[+26.3, +31.1]$ at 20–30 minutes. Comparing the method in seconds instead of whole minutes changes these figures by at most one point.

5.2. Direction of the errors

The two sources fail in opposite directions. The official announcements are late: the median signed error is +124 s at 10–20 minutes and +149 s at 20–30 minutes, and in about half of the announcements made 10 to 30 minutes ahead the bus arrives more than two minutes *earlier* than announced (51–52%, against 16% for the method). This is the costly error for a passenger, who plans to reach the stop shortly before the announced time and finds that the bus has gone. The method leans the other way at long horizons (median −24 s at 10–20 minutes and −60 s at 20–30 minutes): when it errs, the passenger waits a little longer at the stop rather than missing the bus.

The official announcements are not simply the published timetable: only 12–17% of them are within one minute of the timetable time, and the timetable itself is within two minutes of the actual passage in only 14–18% of these cases.

5.3. Days, stops and robustness

The pattern holds on every day (Table 3): the share within two minutes is 33–38% for the official service and 59–72% for the method. It holds on weekdays (37% against 63%) and on Sunday (33% against 72%), and at the priority stops (35% against 62%) as at the others (37% against 66%). For the second announced bus (e_2), which is further away, the official service is within two minutes in 26% of cases and the method in 53%.

Removing the pairs in which another bus of the same line passed within two minutes of the matched one, where the matching could pick the wrong bus, removes 1.3% of the pairs and changes no figure by more than one point.

5.4. The strike day

On Wednesday 11, with minimum services, the official service was within two minutes in 36% of cases and the method in 63%, the same as on a normal day. The method was, however, more often late on that day: in 27% of cases the bus arrived more than two minutes before its prediction, against 12–16% on normal days. Its learned travel times come from normal traffic, and with fewer buses and lighter traffic the buses ran faster than learned.

Table 3: Errors by day, all horizons up to 30 minutes. Columns as in Table 2. Wednesday 11 (general strike) is shown for reference and is not part of the results of the week.

		Official			Method		
Day	Pairs	Median	±2 min	Early	Median	±2 min	Early
2026-11-08	24,820	219	33	55	70	72	16
2026-11-09	31,925	179	38	40	75	67	15
2026-11-10	38,704	186	37	39	84	63	12
2026-11-11 (strike)	24,331	200	36	48	84	63	27
2026-11-12	34,293	179	38	38	84	61	12
2026-11-13	30,304	205	34	41	92	59	16

6. Discussion

6.1. What the comparison shows

On the buses and stops compared, a method that uses only the operator's open data predicts arrivals more accurately than the operator's own service from three minutes before arrival onwards, and the gap is largest, about 34 points in the share within two minutes, when the bus is 6 to 20 minutes away. That is the range in which a passenger decides when to leave for the stop. Below three minutes both are accurate, because both effectively see the bus approaching.

The comparison does not reveal how the official service computes its waiting times, and we do not claim to know. Its errors grow quickly with the horizon and are biased towards announcing the bus later than it comes. That is the behaviour one would expect from a prediction that does not learn how fast buses actually travel on each segment, which is the core of the method (Garrido 2026b).

6.2. Third-party apps

Many passengers do not use the official app but general-purpose journey planners, which receive real-time data from operators. If such an app shows the operator's prediction unchanged, its accuracy is the official accuracy measured here. We have not measured any third-party app directly, and their processing may differ.

6.3. Limitations

- **Which buses are compared.** An announcement can only be compared if the actual passage of the matched bus can be reconstructed from the feed. This excludes buses for which the stop is at or near the

end of their trip (the feed stops reporting them when the trip ends) and buses not reported by the feed. Of the official announcements of the next bus up to 25 minutes ahead, 19% could be paired. The results describe the buses whose passage is observable, which are the majority of buses *in the middle* of their trip, not every announcement.

- **One week, one network.** The data cover five normal days (one of them a Sunday) in November, at 60 stops chosen for their traffic and their difficulty. Other seasons, timetables or networks may differ.
- **Resolution of the actual passage.** The feed changes every 60 seconds, so the actual passage is known to about ±30 s. This adds the same noise to both sources and cannot explain differences of minutes.
- **Matching.** The official service does not identify the vehicle it announces. Matching by order of passage is what a passenger experiences, and the results do not change when ambiguous cases are removed.
- **The method is the replay, not the app.** The method is recomputed from the feed with its reference implementation, with tables frozen before the week. The live app recalibrates every night and also remembers buses that have recently passed, so it may perform slightly differently.

7. Reproducibility

The capture program, the list of stops with the reason for each, the days used (`dias.json`), the comparison (`comparar.py`) and the tables (`resultados.py`) are in the folder `articulo-comparacion` of the repository. The raw captures (feed snapshots, official responses and the timetable of each day) are published as assets of the release `comparacion-2026-11`. The method's reference implementation and the learned tables used are those of the companion paper.

8. Conclusions

For one week, at 60 stops of the Bizkaibus network, we recorded the official real-time waiting times minute by minute and compared them, bus by bus, with the arrivals predicted by an open method that uses only the operator's published positions and timetable. With the bus 6 to 20 minutes away, the official service is within two minutes of the actual passage in about a third of

cases and the method in about two thirds, and the official errors are mostly in the direction that makes passengers miss the bus. Operators that publish their vehicle positions already make possible predictions better than their own; adopting a learned, segment-based method, or opening the predictions to independent evaluation, would directly benefit their passengers.

References

Brakewood, C. and K. Watkins (2019). "A Literature Review of the Passenger Benefits of Real-Time Transit Information". In: *Transport Reviews* 39.3, pp. 327–356. DOI: 10.1080/01441647.2018.1472147.

Cats, O. and G. Loutos (2016). "Real-Time Bus Arrival Information System: An Empirical Evaluation". In: *Journal of Intelligent Transportation Systems* 20.2, pp. 138–151. DOI: 10.1080/15472450.2015.1011638.

Comité Europeo de Normalización (2022). *Public Transport. Service Interface for Real-Time Information Relating to Public Transport Operations. Part 3: Functional Service Interfaces*. Norma europea EN 15531-3:2022.

Efron, B. and R. J. Tibshirani (1993). *An Introduction to the Bootstrap*. New York: Chapman & Hall. ISBN: 978-0-412-04231-7.

García-Mauriño, C., P. J. Zufiria, and A. Jarabo-Peñas (2024). "Improving Bus Arrival Time Predictors Using Only Public Transport API Data". In: *Transportation Letters* 16.8, pp. 804–813. DOI: 10.1080/19427867.2023.2245994.

Garrido, Y. (2026a). *Predicción de llegadas de autobuses con posiciones en tiempo real y horarios interpolados*. Tech. rep. Version 2.1. Zenodo. DOI: 10.5281/zenodo.24041572. In Spanish.

— (Nov. 7, 2026b). *Predicting Bus Arrivals When the Feed Reports Positions but Not Delays: A Reproducible Three-Layer Method with an Open Reference Implementation*. Version 1.0. Preprint. URL: https://github.com/Garridoparrayeray/bide-eta.

Gooze, A., K. E. Watkins, and A. Borning (2013). "Benefits of Real-Time Transit Information and Impacts of Data Accuracy on Rider Experience". In: *Transportation Research Record: Journal of the Transportation Research Board* 2351, pp. 95–103. DOI: 10.3141/2351-11.

Kaufman, S. et al. (2012). "Leakage in Data Mining: Formulation, Detection, and Avoidance". In: *ACM Transactions on Knowledge Discovery from Data* 6.4, 15. DOI: 10.1145/2382577.2382579.

Liu, L. and H. J. Miller (2020). "Does Real-Time Transit Information Reduce Waiting Time? An Empirical Analysis". In: *Transportation Research Part A: Policy and Practice* 141, pp. 167–179. DOI: 10.1016/j.tra.2020.09.014.

Mishalani, R. G., M. M. McCord, and J. Wirtz (2006). "Passenger Wait Time Perceptions at Bus Stops: Empirical Results and Impact on Evaluating Real-Time Bus Arrival Information". In: *Journal of Public Transportation* 9.2, pp. 89–106. DOI: 10.5038/2375-0901.9.2.5.

MobilityData (2026). *GTFS Schedule Reference*. URL: https://gtfs.org/documentation/schedule/reference/ (visited on 10/31/2026).

Rahman, M. M., S. C. Wirasinghe, and L. Kattan (2013). "Users' Views on Current and Future Real-Time Bus Information Systems". In: *Journal of Advanced Transportation* 47.3, pp. 336–354. DOI: 10.1002/atr.1206.

Tashman, L. J. (2000). "Out-of-Sample Tests of Forecasting Accuracy: An Analysis and Review". In: *International Journal of Forecasting* 16.4, pp. 437–450. DOI: 10.1016/S0169-2070(00)00065-0.

Watkins, K. E. et al. (2011). "Where Is My Bus? Impact of Mobile Real-Time Information on the Perceived and Actual Wait Time of Transit Riders". In: *Transportation Research Part A: Policy and Practice* 45.8, pp. 839–848. DOI: 10.1016/j.tra.2011.06.010.

Wessel, N., J. Allen, and S. Farber (2017). "Constructing a Routable Retrospective Transit Timetable from a Real-Time Vehicle Location Feed and GTFS". In: *Journal of Transport Geography* 62, pp. 92–97. DOI: 10.1016/j.jtrangeo.2017.04.012.